Logarithmic and Exponential Functions

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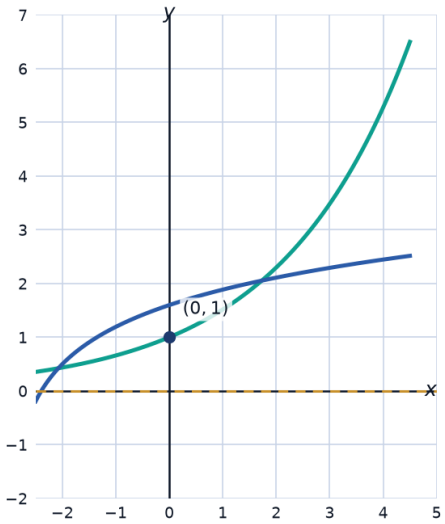
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve the exponential equation $2^{x+2} = 5^x$, giving x in terms of common logarithms.

2 [Foundational] • Target: Fundamental Laws of Logarithms

Solve the exponential equation $2^{x+3} = 5^x$, giving x in terms of common logarithms.



3 [Foundational] • Target: Solving Logarithmic Equations

Solve the exponential equation $2^{x+4} = 5^x$, giving x in terms of common logarithms.

4 [Foundational] • Target: Exponential Equations and Growth Models

Solve the exponential equation $2^{x+5} = 5^x$, giving x in terms of common logarithms.

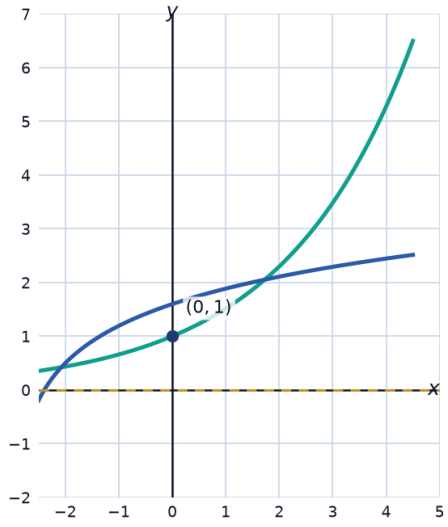
5 [Foundational] • Target: Reduction to Linear Form ($Y = mX + c$)

Solve the exponential equation $2^{x+6} = 5^x$, giving x in terms of common logarithms.

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6 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve the exponential equation $2^{x+7} = 5^x$, giving x in terms of common logarithms.



7 [Foundational] • Target: Fundamental Laws of Logarithms

Solve the exponential equation $2^{x+8} = 5^x$, giving x in terms of common logarithms.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving Logarithmic Equations

The real numbers x and y satisfy the simultaneous equations: $\log_3(x) + 2\log_9(y-1) = 2$ $4^y \cdot 8^{-\frac{x}{3}} = 32$

Find the value of $x^2 + y^2$.

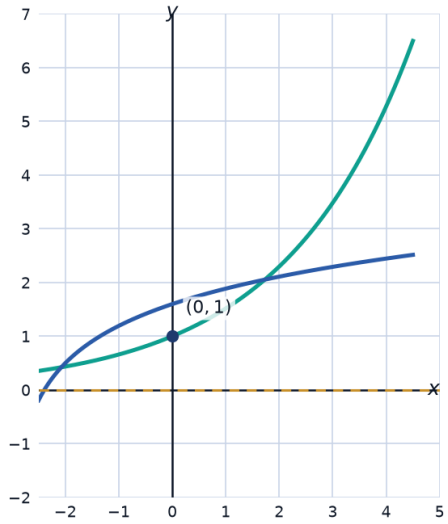
9 [Intermediate] • Target: Exponential Equations and Growth Models

The variables x and y, where $x > 0$, satisfy the relationship $y = ax + \frac{b}{x}$ where a and b are positive constants. When xy is plotted on the vertical axis against x^2 on the horizontal axis, a straight line is obtained which passes through the points $(2, 11)$ and $(5, 23)$. Given further that x and y satisfy the logarithmic equation $\log_3(y - ax) + \log_9(x^2y^2) = 1 + \log_3(x + 10)$ find the sum of all possible values of x.

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10 [Intermediate] • Target: Reduction to Linear Form ($Y = mX + c$)

Solve the logarithmic equation $\log_2(5x + 6) - \log_2(x) = 3$.



11 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 3$. What is the value of x? Show your complete algebraic work.

12 [Intermediate] • Target: Fundamental Laws of Logarithms

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 4$. What is the value of x? Show your complete algebraic work.

13 [Intermediate] • Target: Solving Logarithmic Equations

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 5$. What is the value of x? Show your complete algebraic work.

14 [Intermediate] • Target: Exponential Equations and Growth Models

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 6$. What is the value of x? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

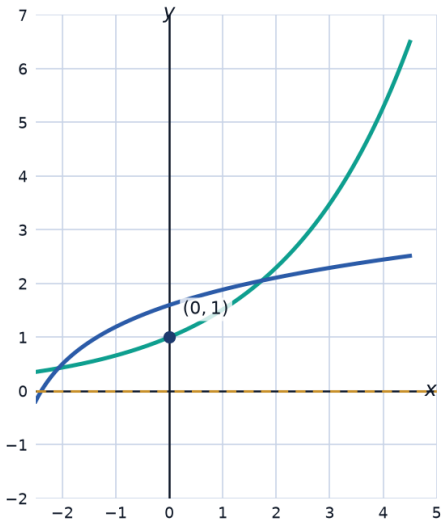
15 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 7$. What is the value of x? Show your complete algebraic work.

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16 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 8$. What is the value of x? Show your complete algebraic work.



17 [Challenging] • Target: Fundamental Laws of Logarithms • Cross-Lesson: 7.01 Differentiation and Rates of Change

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 9$. What is the value of x? Show your complete algebraic work.

18 [Challenging] • Target: Solving Logarithmic Equations • Cross-Lesson: 8.01 Integration and Kinematics

Variables x and y satisfy the equation $y = Ax^k$, where A and k are constants, with $x > 0$. The graph of $\ln y$ against $\ln x$ is a straight line passing through the points $(\ln 2, \ln 72)$ and $(\ln 6, \ln 8)$. Find the value of x when $y = 18$.

19 [Challenging] • Target: Exponential Equations and Growth Models • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The variables x and y satisfy the relationship $y = Ax^n$, where A and n are constants. When $\log_2 y$ is plotted against $\log_2 x$, a straight line is obtained which passes through the points $(1, 5)$ and $(4, -1)$. What is the value of y when $x = 4$? Show your complete algebraic work.

20 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

The variables x and y satisfy the equation $y = Ab^x$, where A and b are positive constants. When $\log_2 y$ is plotted against x, a straight line is obtained that passes through the points $(2, 7)$ and $(5, 13)$. Find the value of y when $x = 4$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{3}(x) + \log_{3}(4) = \log_{3}(16)$.

a. 4

b. 6

c. 8

d. 2

22 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{4}(x) + \log_{4}(4) = \log_{4}(36)$.

a. 3

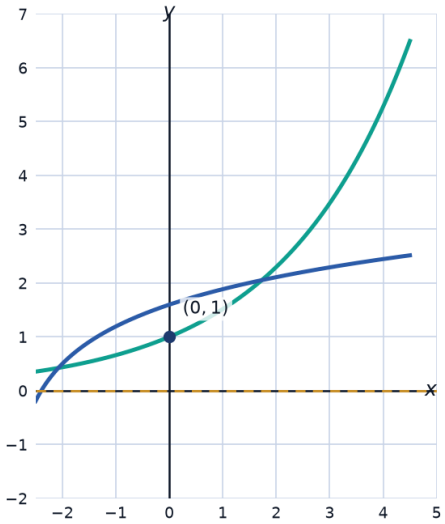
b. 18

c. 11

d. 9

23 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{5}(x) + \log_{5}(4) = \log_{5}(64)$.



a. 4

b. 32

c. 18

d. 16

24 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{6}(x) + \log_{6}(4) = \log_{6}(100)$.

a. 25

b. 27

c. 5

d. 50

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25 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{7}(x) + \log_{7}(4) = \log_{7}(144)$.

a. 38

b. 36

c. 72

d. 6

26 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{8}(x) + \log_{8}(4) = \log_{8}(196)$.

a. 7

b. 98

c. 49

d. 51

27 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{9}(x) + \log_{9}(4) = \log_{9}(256)$.

a. 66

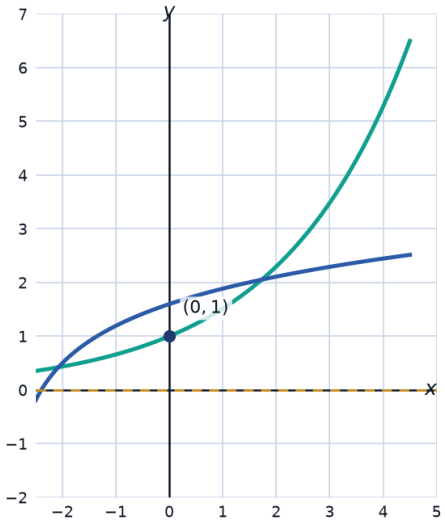
b. 128

c. 64

d. 8

28 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{10}(x) + \log_{10}(4) = \log_{10}(324)$.



a. 9

b. 162

c. 83

d. 81

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Fundamental Laws of Logarithms**

Evaluate $\log_3(54) - \log_3(2)$ without a calculator.

a. 3

b. 2

c. 4

d. $\log_3(52)$

30 **[Intermediate]** • **Target: Fundamental Laws of Logarithms**

Express $2\log_a(x) - 3\log_a(y) + \log_a(z)$ as a single logarithm.

a. $\log_a\left(\frac{x^2z}{y^3}\right)$

b. $\log_a\left(\frac{2xz}{3y}\right)$

c. $\log_a(x^2 - y^3 + z)$

d. $\log_a\left(\frac{x^2}{y^3z}\right)$

31 **[Intermediate]** • **Target: Fundamental Laws of Logarithms**

Given $\log_b(2) = p$ and $\log_b(3) = q$, express $\log_b(72)$ in terms of p and q.

a. $3p + 2q$

b. $2p + 3q$

c. $6pq$

d. $3p - 2q$

32 **[Intermediate]** • **Target: Solving Logarithmic Equations**

Solve the equation $\log_2(x) + \log_2(x - 2) = 3$ for $x > 2$.

a. $x = 4$

b. $x = 4$ and $x = -2$

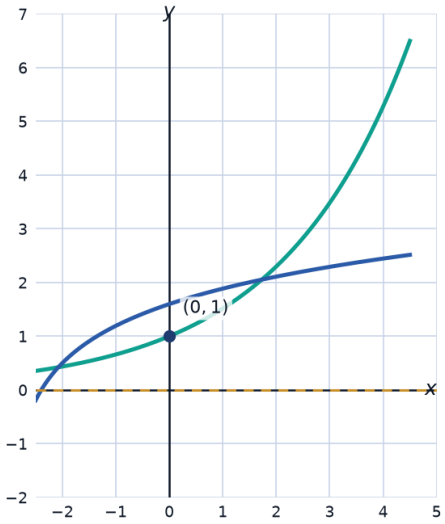
c. $x = 3$

d. $x = 8$

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33 [Intermediate] • Target: Solving Logarithmic Equations

Solve $\log_3(2x+1) - \log_3(x-1) = 2$.



a. $x = \frac{10}{7}$

b. $x = 2$

c. $x = \frac{7}{10}$

d. No real solution

34 [Intermediate] • Target: Solving Logarithmic Equations

Solve the base-change logarithmic equation $\log_4(x) + \log_2(x) = 6$.

a. $x = 16$

b. $x = 8$

c. $x = 64$

d. $x = 4$

35 [Intermediate] • Target: Exponential Equations and Growth Models

Solve $3^{2x-1} = 5^x$, giving x in terms of natural logarithms.

a. $x = \frac{\ln 3}{2\ln 3 - \ln 5}$

b. $x = \frac{\ln 5}{\ln 3}$

c. $x = \frac{\ln 3}{\ln 5 - 2\ln 3}$

d. $x = \frac{2\ln 3}{\ln 5}$

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36 [Intermediate] • Target: Exponential Equations and Growth Models

A bacterial culture grows according to $N(t) = 500e^{0.04t}$, where t is in hours. After how many hours will the population reach 2000?

a. $t = 25\ln 4 \approx 34.7$ hr

b. $t = 50$ hr

c. $t = 20\ln 2 \approx 13.9$ hr

d. $t = 100$ hr

37 [Intermediate] • Target: Exponential Equations and Growth Models

Solve the simultaneous exponential system $2^x 4^y = 32$ and $3^{x-y} = 27$.

a. $x = 4, y = \frac{1}{2}$

b. $x = 3, y = 1$

c. $x = 5, y = 2$

d. $x = 2, y = 1$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 8.01 Integration and Kinematics

The variables x and y are related by $y = ax^b$. When $\ln y$ is plotted against $\ln x$, a straight line passes through $(0, 2)$ with slope 3. Find a and b.

a. $a = e^2, \quad b = 3$

b. $a = 2, \quad b = 3$

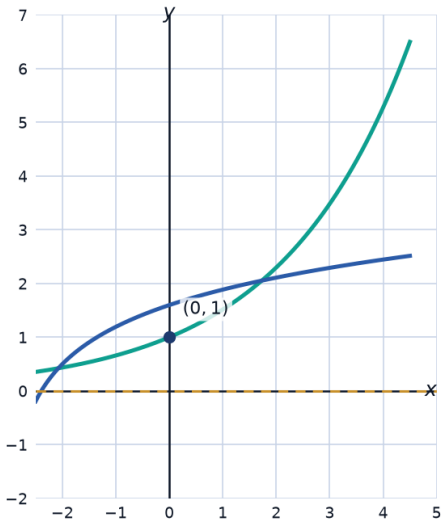
c. $a = 3, \quad b = 2$

d. $a = e^3, \quad b = e^2$

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39 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The relation $y = Ab^x$ is linearized by plotting $\lg y$ against x. If the straight line has y-intercept 1 and gradient 0.3, find A and b.



a. $A = 10, \quad b = 10^{0.3}$

b. $A = 1, \quad b = 0.3$

c. $A = 10^{0.3}, \quad b = 10$

d. $A = 0.3, \quad b = 10$

40 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

If a graph of xy against x^2 is a straight line with slope 4 and vertical intercept -5, express y in terms of x.

a. $y = 4x - \frac{5}{x}$

b. $y = 4x^2 - 5$

c. $y = 4 - \frac{5}{x^2}$

d. $y = \frac{4}{x} - 5$

41 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Find the real solution to the equation $\log_3 x + \log_3 (x - 8) = 2$.

a. $x = 9$

b. $x = 9$ and $x = -1$

c. $x = 8.5$

d. $x = 5$

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42 [Challenging] • Target: Fundamental Laws of Logarithms • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The variables x and y satisfy the relationship $y = Ab^x$, where A and b are positive constants. The graph of $\ln y$ against x is a straight line passing through the points $(0, \ln 3)$ and $(4, \ln 48)$. What is the value of b?

a. 2

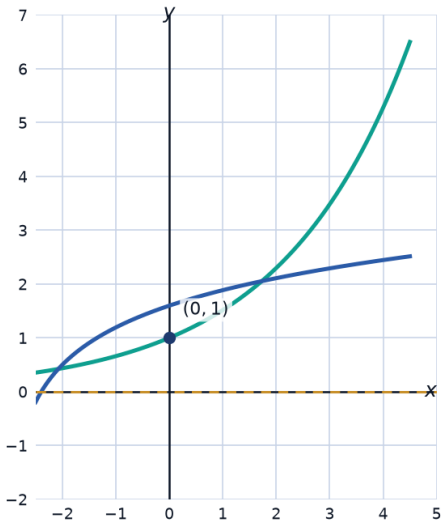
b. 4

c. 16

d. $\sqrt{2}$

43 [Challenging] • Target: Solving Logarithmic Equations • Cross-Lesson: 6.01 Permutations and Combinations

Solve $e^{2x} - 23e^x + 130 = 0$ for x. Give the larger solution in exact form.



a. $\ln 10$

b. $\ln 13$

c. $\ln 23$

d. $\ln 130$

44 [Challenging] • Target: Exponential Equations and Growth Models • Cross-Lesson: 7.01 Differentiation and Rates of Change

Solve $e^{2x} - 21e^x + 110 = 0$ for x. Give the larger solution in exact form.

a. $\ln 11$

b. $\ln 110$

c. $\ln 21$

d. $\ln 10$

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45 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 8.01 Integration and Kinematics

Solve $e^{2x} - 28e^{x} + 187 = 0$ for x. Give the larger solution in exact form.

a. $\ln 187$

b. $\ln 17$

c. $\ln 11$

d. $\ln 28$

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